

QBUS5001

Q&A Lecture notes - Week 11

Part (-1) – Administrative matters (Same as last week)

0.1 Group assignment is available now

Your group assignment will be available on 2:05pm 08/10/2024 via the link:

<https://canvas.sydney.edu.au/courses/61229/assignments/549695>

0.2 Nomination for Students' Choice Award for Teaching (Deadline 1 November 2024)

If you've found my teaching helpful, I would greatly appreciate your nomination for the Students' Choice Award for Teaching. This would not only benefit me but also help show others how unique and engaging a university classroom can be.

The nomination link is attached below:

<https://forms-business.sydney.edu.au/teaching/students-choice-award-for-teaching>

Here are the details you need to submit your nomination:

Nominee details		
Are you nominating an individual or a team?*		
☒ Individual ☐ Team		
Are you nominating this individual/team based on their achievements in S2 2024 incl. intensives or S1 2025 incl. intensives, or both?*		
☒ S2 2024 incl. intensives ☐ S1 2025 incl. intensives		
Who are you nominating? Please ensure that they have agreed to be nominated.		
First name*	Last name*	University email*
Ce (Tony)	Shang	ce.shang@sydney.edu.au
What is the relevant Discipline/Program for this individual or team?*		
Business Analytics		

0.3 Nomination for 'Wayne Lonergan Award for Outstanding Leadership in Teaching' (Deadline 1 November 2024)

There's another award, the *Wayne Lonergan Award for Outstanding Leadership in Teaching*, which also aligns with my work. It seems more complex, but if you feel I'm deserving, I'd love to give it a shot!

Here's the nomination link:

<https://forms-business.sydney.edu.au/teaching/WLOT-nominations>

The same details are needed for the nomination:

Nominee details		
Are you nominating an individual or a team?*		
☒ Individual ☐ Team		
Are you nominating this individual/team based on their achievements in S2 2024 incl. intensives or S1 2025 incl. intensives, or both?*		
☒ S2 2024 incl. intensives ☐ S1 2025 incl. intensives		
Who are you nominating? Please ensure that they have agreed to be nominated.		
First name*	Last name*	University email*
Ce (Tony)	Shang	ce.shang@sydney.edu.au
What is the relevant Discipline/Program for this individual or team?*		
Business Analytics		

I truly appreciate your time and effort in the nomination process. Thank you in advance!

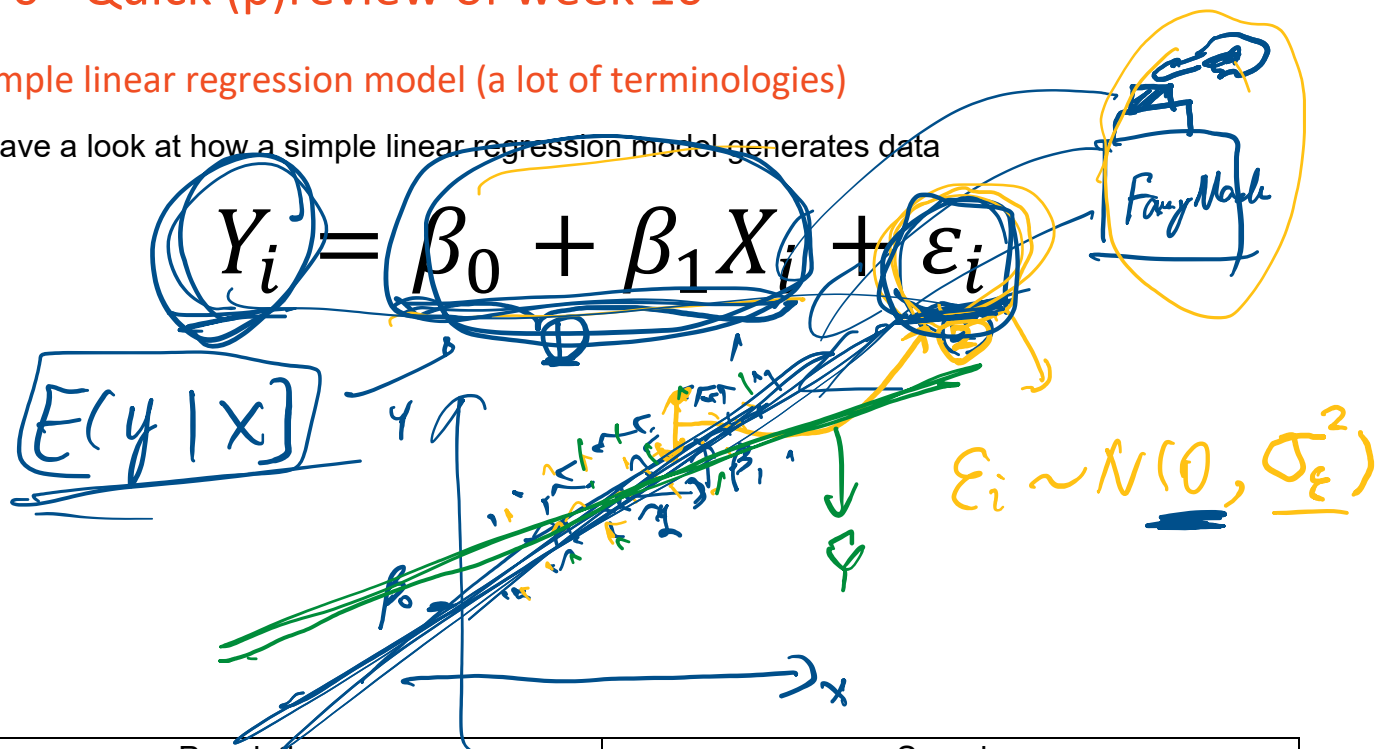
I truly believe we are witnessing history together! I never imagined things would unfold in such an incredible way, and none of this would be possible without your engagement and support. We've made a huge impact, not just within our classroom, but beyond it as well.

Together, we are making a real difference!

Part 0 - Quick (p)review of week 10

1.2 Simple linear regression model (a lot of terminologies)

Let's have a look at how a simple linear regression model generates data



Population	Sample
Population regression equation $E(y x) = \beta_0 + \beta_1 X$	Estimated regression equation $\hat{Y} = b_0 + b_1 X$ $\hat{\beta}_0 + \hat{\beta}_1 X$
Population slope coefficient β_1	Sample slope coefficient b_1 or $\hat{\beta}_1$
Population intercept β_0	Sample intercept b_0 or $\hat{\beta}_0$
Population error term $\epsilon_i = Y_i - E(Y_i X_i)$	Sample residual $e_i = Y_i - \hat{Y}_i$

Part 1 – Residual analysis: checking model assumptions

ϵ_i

Last week, in terms of the population error term, we've made some assumptions on the error term:

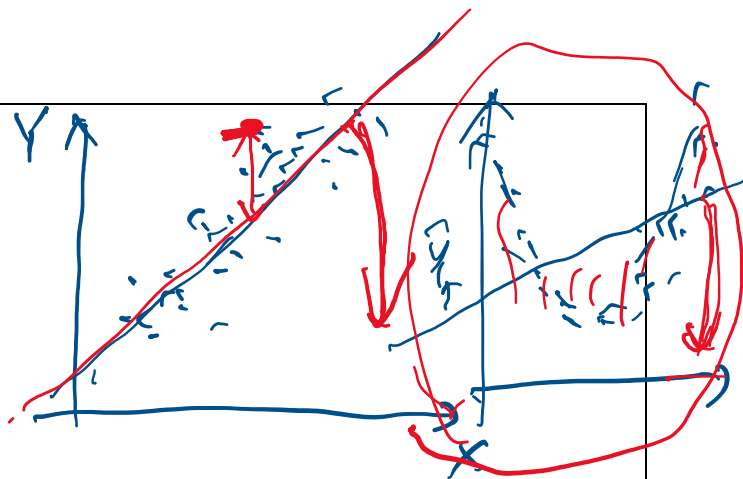
$$\varepsilon_i \sim N(0, \sigma_\varepsilon^2) \xrightarrow{i.i.d} \text{independent \& identically distributed}$$

Now we can break it down into four assumptions to help you better memorise them, those are:

L.I.N.E

Assumption 1 – Linearity

L.I.N.E



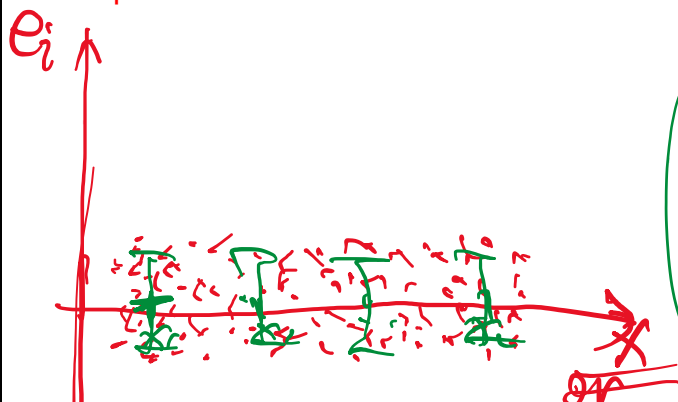
Requirement:

$$E[\varepsilon_i | X] = 0$$

How can we check the validity of this assumption?

residual plot

Assumption satisfied



Assumption violated



How to fix the model when the assumption is violated?

$$y = \beta_0 + \beta_1 x^2 + \varepsilon_i$$

transformations

$$y = \ln x$$

Assumption 2 – Independence of errors

L.I.N.E

2

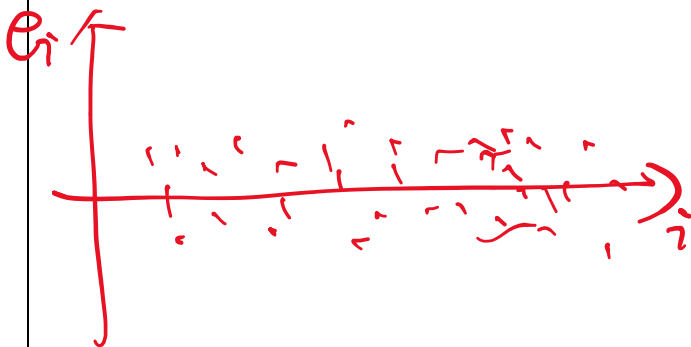
Requirement:

$$\text{Corr}(\varepsilon_i, \varepsilon_j) = 0 \rightarrow \text{usually relevant when you have a time-series data}$$

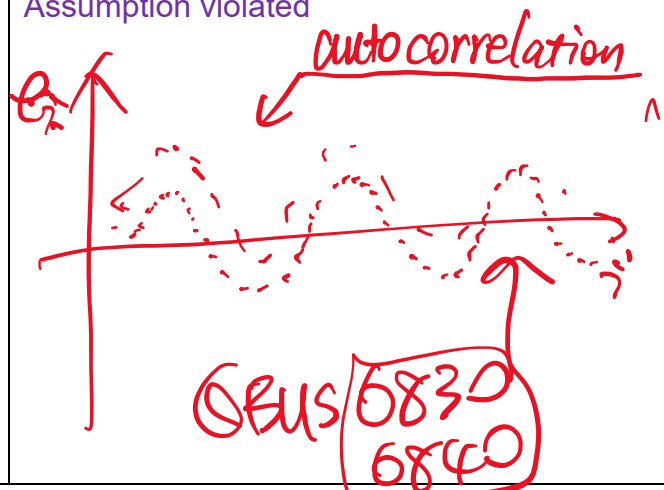
How can we check the validity of this assumption?

1. residual plot $\Rightarrow$ ~~e_i~~ against $\hat{y}_i$

Assumption satisfied



Assumption violated



2. Durbin - Watson test $\Rightarrow$ self-study

How to fix the model when the assumption is violated?

$$\textcircled{y_i} = \beta_0 + \beta_1 X_i + \textcircled{\beta_2 y_{i-1}} + \varepsilon$$

Assumption 3 – Normality of errors

L.I.N.E

$$\varepsilon_i \sim N$$

Requirement:

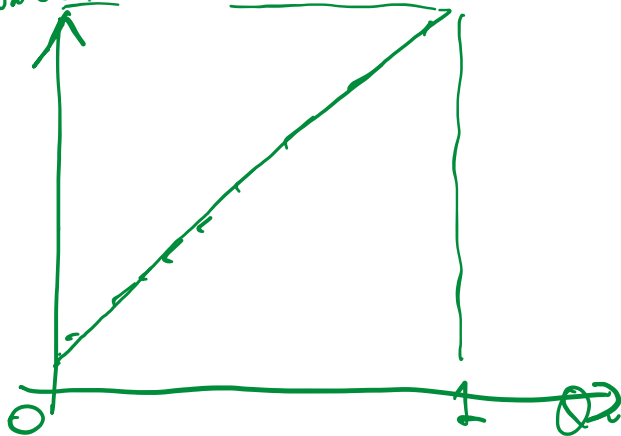
$$\varepsilon_i \sim N$$

How can we check the validity of this assumption?

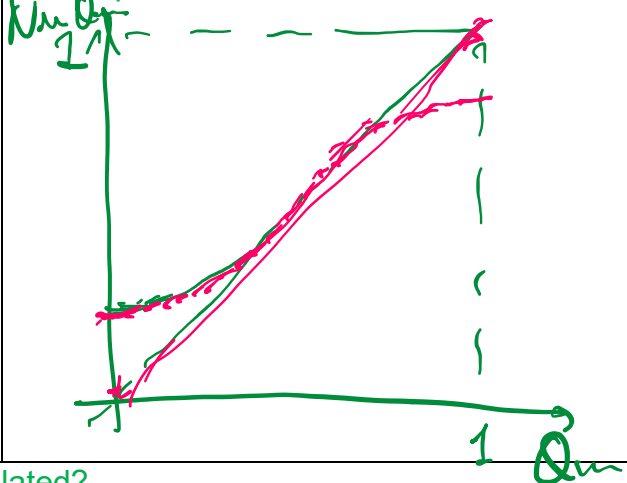
QQ - plot
Quantile - Quantile

Note: if $n \geq 30$, CLT applies.
Normality Assumption is not that essential

Assumption satisfied



Assumption violated



How to fix the model when the assumption is violated?

learn in the future

econometrics

Assumption 4 – Equal variance (Homoscedasticity)

L.I.N.E

$$SER = \sqrt{\frac{SSE}{n-2}}$$

Requirement:

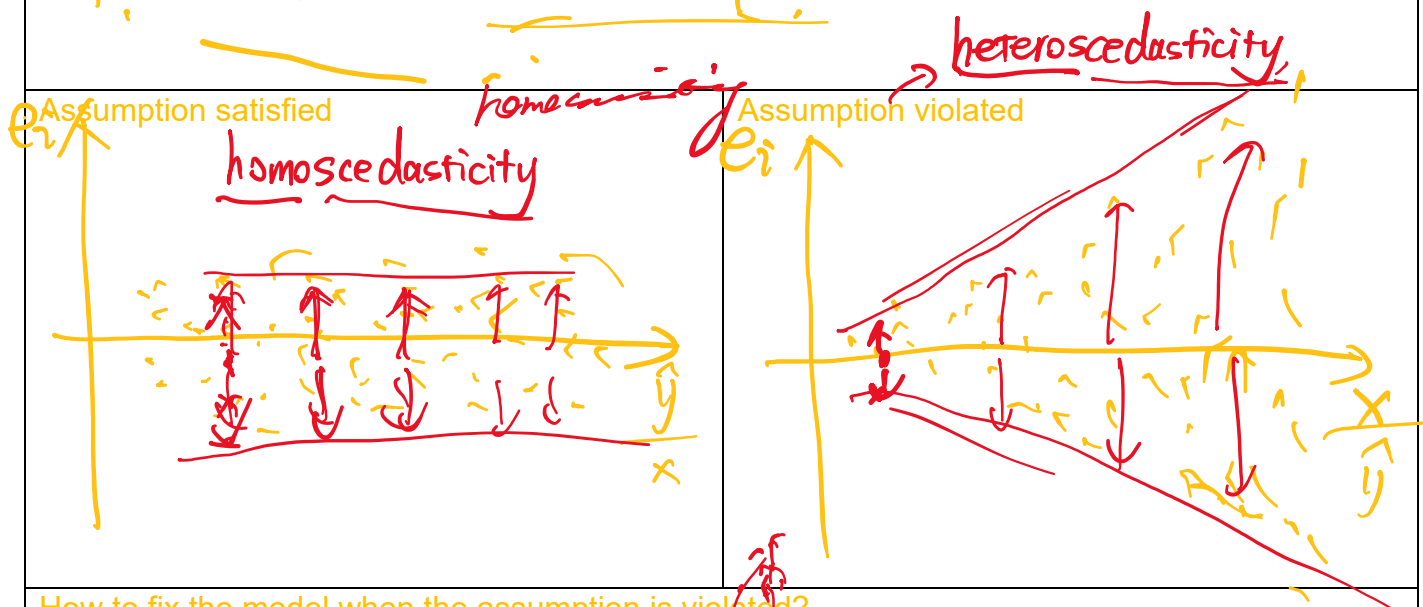
$$\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$$

$$V[\varepsilon_i | X] = \sigma_\varepsilon^2$$

$X_1, X_2, X, \dots$

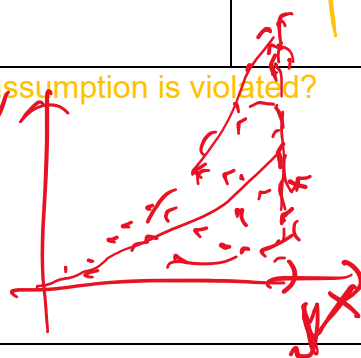
How can we check the validity of this assumption?

Residual vs X or $\hat{y}$.



How to fix the model when the assumption is violated?

transformation →
log transformation



Example Question

A research study is investigating the phenomenon of fire alarm fatigue, where frequent fire alarms cause individuals to become desensitised, reducing their responsiveness to real emergencies. The study aims to predict the response time to a fire alarm (Y) based on the number of false alarms experienced (X) in a given month.

The data is stored in "QBUS5001 Week 11 Lecture Data.xlsx".

Number of False Alarms (X)	Average Response Time (seconds) (Y)
7	53.29
15	66.58
11	64.63
8	53.59
7	53.34
19	90.68
11	59.1
11	58.78
4	41.25
8	49.17
3	39.75
2	35.43
12	79.17
6	42.87
2	32.09
1	35.12
12	70.36

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.953436					
R Square	0.90904					
Adjusted R Square	0.908112					
Standard Error	5.000925					
Observations	100					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	1	24493.92	24493.92	979.3943	8.11E-53	
Residual	98	2450.906	25.00925			
Total	99	26944.82				
Coefficients						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	30.55761	1.042588	29.30937	2.71E-50	28.48863	32.6266
Number	2.969874	0.094898	31.29528	8.11E-53	2.781551	3.158197

$SER = \hat{\sigma}_{\epsilon}$

Questions:

- Using the least squares method, estimate the simple linear regression equation that predicts the response time (Y) based on the number of false alarms (X).

$$\hat{Y} = 30.56 + 2.97X$$

$b_1 = 2.97$

$\beta \neq 0$

- Perform a residual analysis for these data. Evaluate whether the assumptions of regression have been seriously violated.

Linearity
Independence of error
Normality of errors
Equal variance

Part 2 – Hypothesis test of the linear relationship

In regression analysis, two key questions often arise: **How strong is the linear relationship between two variables?** and **Is the observed relationship likely to be genuine or just due to chance?** These questions are fundamental in understanding the reliability and significance of any linear association between variables.

To assess the strength and significance of a linear relationship, two primary statistical tests can be applied:

1. **Testing the Correlation Coefficient (ρ):** This test evaluates the strength and direction of the linear relationship between two variables. The correlation coefficient, ρ , ranges from -1 to 1, where values close to 1 or -1 indicate a strong positive or negative linear relationship, respectively, and values near 0 suggest no linear relationship. Hypothesis testing on ρ determines whether the observed correlation is statistically significant or could have occurred by random chance.
2. **Testing the Regression Slope Coefficient (β):** The slope of the regression line, β , represents the rate of change in the dependent variable for every one-unit change in the independent variable. By testing β , we can determine whether there is a statistically significant linear relationship between the variables. If β is significantly different from zero, we conclude that changes in the independent variable are associated with changes in the dependent variable and that this relationship is unlikely due to chance.

Test #1 Testing the correlation coefficient ρ

The null hypothesis is

$$H_0: \rho = \rho_0$$

And the alternative hypothesis is determined by whether it is a one-tailed or two-tailed test.

Under H_0 the test statistic is given by

$$T = \frac{r - \rho_0}{\sqrt{\frac{1 - r^2}{n - 2}}} \sim t_{n-2}$$

If the null hypothesis is rejected, it indicates that there is sufficient evidence to support the existence of a strong linear relationship between the two variables.

Handwritten notes and diagrams:

- Diagram showing the relationship between population correlation (ρ) and sample correlation (r). Arrows point from ρ to r and vice versa, with a central arrow pointing to the hypothesis testing process.
- Labels: "population corr" next to ρ , "sample corr" next to r .
- Text: "Hypothesised population correlation coefficient" with an arrow pointing to the hypothesis testing section.
- Hypotheses listed:
 - $H_0: \rho = 0$ (crossed out)
 - $H_1: \rho \neq 0$
 - $H_1: \rho > 0$
 - $H_1: \rho < 0$

Test #2 Testing the regression slope coefficient β_1

The null hypothesis is

$$H_0: \beta_1 = \text{hypothesised } \beta_1$$

And the alternative hypothesis is determined by whether it is a one-tailed or two-tailed test.

Under H_0 the test statistic is given by

$$T = \frac{b_1 - \beta_1}{S_{b_1}} \sim t_{n-2}$$

Where

$$S_{b_1} = \frac{SER}{\sqrt{SSX}} = \frac{SER}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

If the null hypothesis is rejected, it indicates that there is sufficient evidence to support the existence of a strong linear relationship between the two variables.

Also, the confidence interval of β_1 is given by

$$b_1 \pm t_{\frac{\alpha}{2}, n-2} \times S_{b_1}$$

Example Question (Continued)

A research study is investigating the phenomenon of **fire alarm fatigue**, where frequent fire alarms cause individuals to become desensitised, reducing their responsiveness to real emergencies. The study aims to predict the **response time to a fire alarm (Y)** based on the **number of false alarms experienced (X)** in a given month.

Questions:

- At 5% level of significance, is there evidence of a linear relationship between response time and number of false alarms experienced.

Now, open the data file titled "**QBUS5001 Week 11 Lecture Data.xlsx**" and navigate to the sheet labeled "**2 Hypothesis Test.**"

Let's review the test and explore how conclusions are drawn based on the results.

Good news
Excel → output

$$H_0: \beta_1 = 0$$

$$\text{vs } H_1: \beta_1 \neq 0$$

← given in Excel.

Part 3 – Confidence interval and prediction interval

Let's continue with the example:

A research study is investigating the phenomenon of fire alarm fatigue, where frequent fire alarms cause individuals to become desensitised, reducing their responsiveness to real emergencies. The study aims to predict the response time to a fire alarm (Y) based on the number of false alarms experienced (X) in a given month.

And in question 1, you've got an estimated regression equation below:

$$\hat{Y} = 30.56 + 2.97X$$

$$\bar{X} = 1.39$$
$$\pm (t) \times SE$$

Questions:

4. Predict the response time if a person experiences 10 false alarms in a month.

$$\hat{Y} = 30.56 + 2.97 \times 10$$
$$= 60.27$$

The calculation is easy, but is it good enough?

No

Confidence interval vs prediction interval

A **confidence interval** provides a range of values that is likely to contain the **true mean** of the dependent variable for a given value of the independent variable. For example, a 95% confidence interval means that if we were to sample data repeatedly, 95% of the time, the interval would contain the true population mean. Confidence intervals are typically narrower because they focus on estimating the **average** value of the response variable for a given input.

A **prediction interval** on the other hand, is used to predict the value of the dependent variable for an **individual** observation at a given value of the independent variable. It accounts not only for the uncertainty in estimating the mean but also for the variability in individual data points. Therefore, prediction intervals are wider, as they reflect the potential spread of individual outcomes around the mean.

$$\bar{y} \sim N(\mu, \sigma^2/n)$$

$$SE(\bar{y}) = \sigma / \sqrt{n}$$

What's the formula? (WTF?)

$$\hat{y} \pm t_{\alpha/2, n-2} \times SER \times \sqrt{h_i}$$

Where

$$h_i = \frac{1}{n} + \frac{(X_i - \bar{X})^2}{SSX}$$

What's the formula? (WTF?)

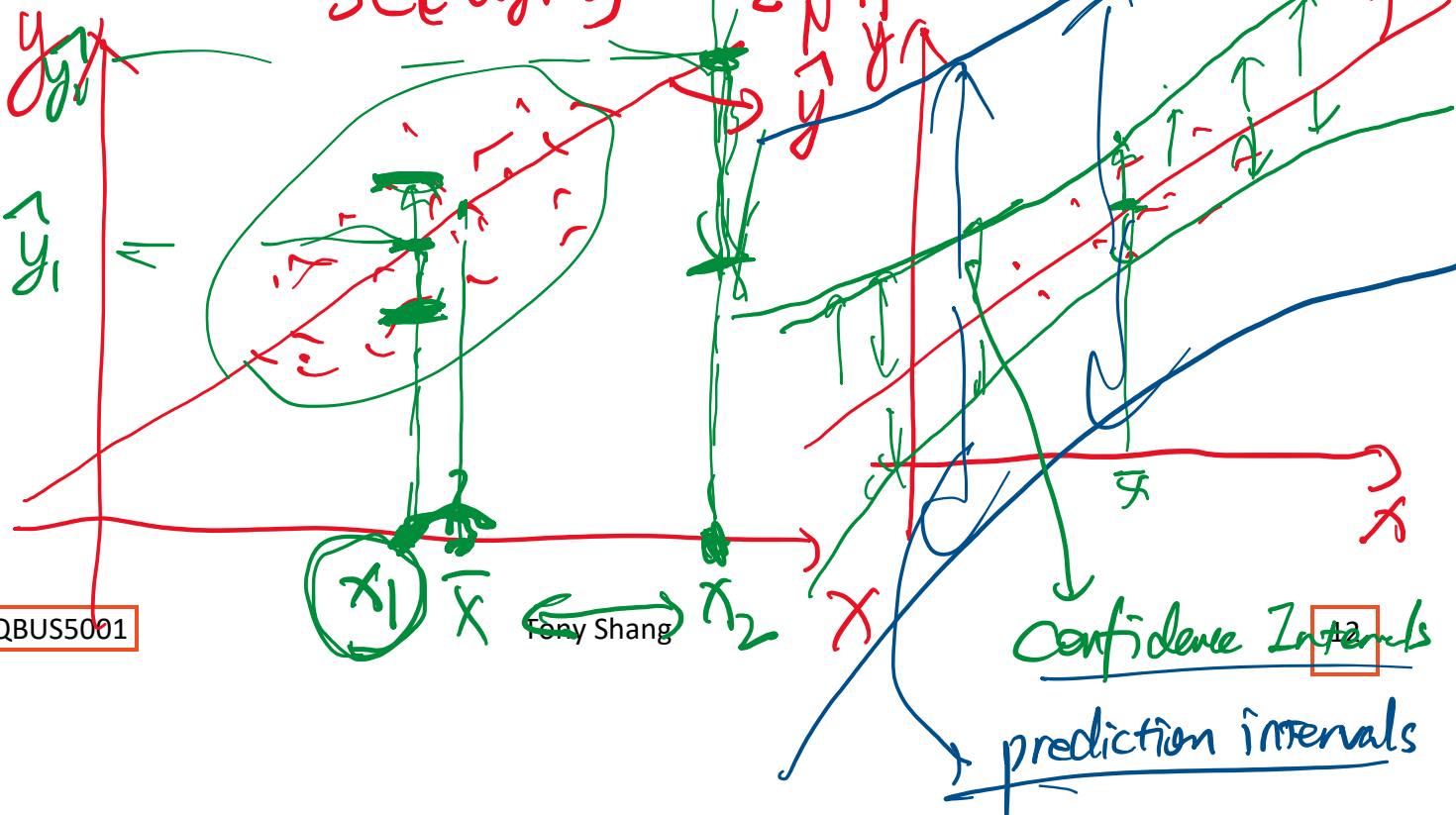
$$\hat{y} \pm t_{\alpha/2, n-2} \times SER \times \sqrt{1 + h_i}$$

Where

$$h_i = \frac{1}{n} + \frac{(X_i - \bar{X})^2}{SSX}$$

Understanding the formula

$$SE(\hat{y}(x)) = \sigma_{\epsilon} \sqrt{\frac{1}{n} + \dots}$$



Now let's quickly apply it!

Example Question (Continued)

A research study is investigating the phenomenon of **fire alarm fatigue**, where frequent fire alarms cause individuals to become desensitised, reducing their responsiveness to real emergencies. The study aims to predict the **response time to a fire alarm (Y)** based on the **number of false alarms experienced (X)** in a given month.

And in question 1, you've got an estimated regression equation below:

$$\hat{Y} = 30.56 + 2.97X$$

And in question 4, you predicted the response time if a person experiences 10 false alarms in a month.

$$\hat{Y}_{|X=10} = 60.26$$

Now, consider the following scenarios; should you use confidence interval or prediction interval?

5. Predict the **average response time** for all individuals who experienced 10 false alarms in a month.

CI

6. Predict the **response time** for Tony, who experienced 10 false alarms in a month.

PI

Once you have answered these questions, proceed to the Excel data file to complete the calculations.